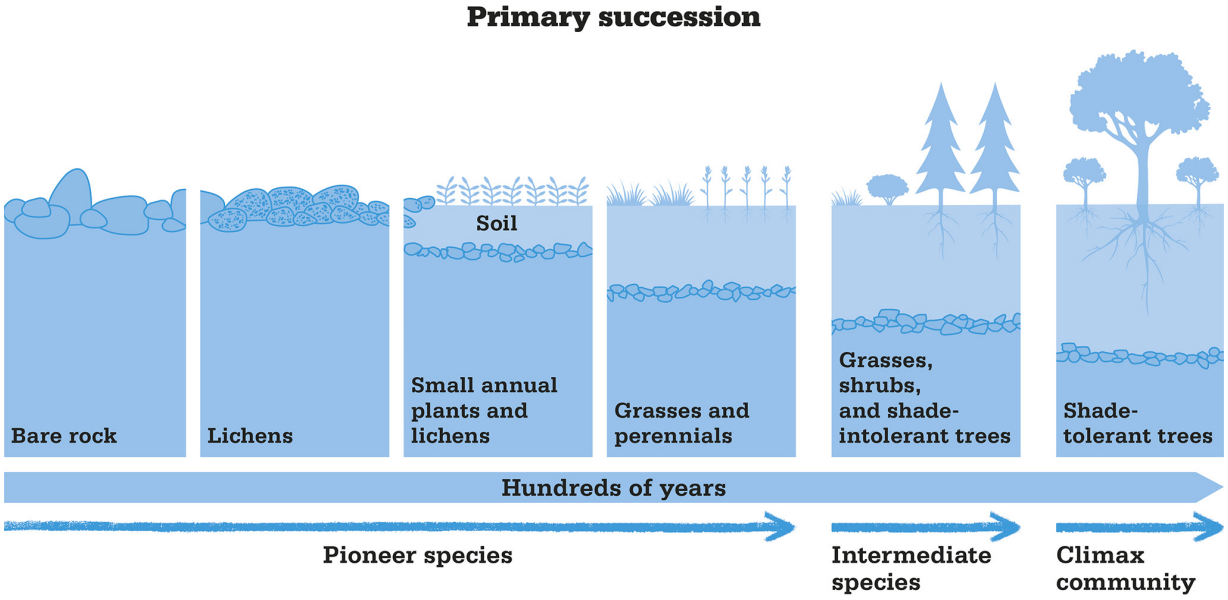
Growth of an ecosystem

French geographer Adolphe Dureau de la Malle is regarded as the first person to use the term “succession” with reference to ecology when he witnessed the progression of plant communities after all the trees were removed from a forest. Cowles provided a more formal articulation of his ecological succession theory, in *The Ecological Relations of the Vegetation on the Sand Dunes of Lake Michigan*, published in 1899. In this seminal paper, he proposed the idea of primary succession—the gradual growth of an ecosystem originally largely devoid of plant life. The stages of primary succession include pioneer plants (often lichens and mosses), followed by grassy plants, small shrubs, and trees.



The process of primary succession begins in barren environments such as bare rock. Hardy species, usually lichens, appear first and then give way to a stable climax community of more complex and diverse life forms over hundreds of years.

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